

SCARLETT CHEN

High school student interested in
electrical engineering

Edgemont Junior-Senior High School, Scarsdale NY

SUMMARY

Hi! I am a prospective electrical engineer (and design minor!), who is especially interested in power electronics, signal processing, and renewable energy. Here are some of the things I've done!

EDUCATION

Skills

- 7 years of **programming experience**
 - Python, Java, and R
 - Data Science & Machine Learning basics
- **Arduino, electronics, & soldering**

Relevant Academics

- **MIT BWSI Basics of ASICs (July 2026)**
 - Computer engineering fundamentals
 - ASIC design; Verilog, OpenLane flow, etc.
- **AP Physics C: Mechanics** (scored 5, 2026)
- **AP Physics C: Electricity & Magnetism** (scored 5, self-study, 2026)
- **AP Calculus AB** (scored 5, self-study, 2026)
- **Linear Algebra & Mathematics of Machine Learning** (unofficially taught by previous mentor, 2025)
- **Data Structures and Algorithms** (Johns Hopkins CTY, 2023)
- **AP Computer Science Principles** (scored 5, self-study, 2025)
- **Current:** AP Calculus BC, AP Chemistry

OTHER

- **Yearbook Director of Photography** (2025-Present)
 - Print design, organization, and leadership
- **Gold Presidential Medal of Service:** 100 hours of volunteer work as a junior instructor at my martial arts school (2023-2024)
- Dartmouth Alumni Book Award (2026)
- 1600 SAT / 1510 PSAT (2025); National Merit Semifinalist (2026)

EXPERIENCE

Research in Power Grid Modernization

2026

- Research (mentored) into the **inertia measurement filtering-lag tradeoff applied to virtual synchronous machines**
 - i.e. improving grid stability for high renewable energy
- Applying for Regeneron STS, aiming for publication in IEEE conference or journal

Research in Computational Biology

2025

- **Team research** (mentored) based in CT
- *Wavelet-Based Multiscale Anisotropy-Aware Cell-Cell Communication and Niche Segmentation*
- Presented & published in ICCBB ([ACM linked here](#))
- Connecticut Science and Engineering Fair Awards
 - **1st Place – Computational Biology and Bioinformatics**
 - **3rd Place – Life Sciences Team**

Science Olympiad

2024-Present -- 2026 Results (Regional)

- **Robot Tour: 1st place:** Build a robot to navigate a walled maze
- **Circuit Lab: 2nd place:** Test on electricity & magnetism and circuits along with a practical lab exam
- **Electric Vehicle: 5th place:** Built a [battery-powered robot](#); top team to do so from scratch (without a kit)

Girls Who Code Leader

2022-Present

- Revived the club from being the only member to ~10 members
- Preparing lesson plans and teaching concepts
- Set up a **collaboration with Edgemont Day Camp**; taught incoming 3rd graders basic computer concepts for ~5 weeks